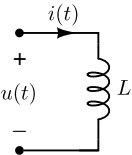


$$u(t) = \sqrt{2}U \cos(\omega t + \varphi_u) = \Re[\sqrt{2}e^{j\omega t}\mathcal{U}]$$

$$i(t) = \sqrt{2}I \cos(\omega t + \varphi_i) = \Re[\sqrt{2}e^{j\omega t}\mathcal{I}]$$



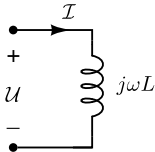
$$u(t) = L \cdot \frac{di(t)}{dt}$$

$$\Downarrow$$

$$\Re[\sqrt{2}e^{j\omega t}\mathcal{U}] = \Re[\sqrt{2} \cdot j\omega e^{j\omega t} L\mathcal{I}]$$

$$\Downarrow$$

$$\Re[\sqrt{2}e^{j\omega t} \{\mathcal{U} - j\omega L \cdot \mathcal{I}\}] = 0 \quad \Rightarrow$$



$$\boxed{\mathcal{U} = j\omega L \cdot \mathcal{I}}$$